Local and Global Identifiability Can Differ in Linear Networks

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***Abstract*— For a linear network, the network matrix collects complex coefficients that relate its node signals. Local identifiability means that the observed response to external inputs uniquely determines all sufficiently close admissible coefficient vectors; global identifiability requires uniqueness among all admissible vectors. For scalar coefficients, generic means that exceptions have zero volume in parameter space. We show that generic local identifiability does not imply generic global identifiability when only selected nodes receive inputs and are measured. A six-node network with nine independent unknown coefficients has exactly two compatible vectors generically. Adding six nodes, each supplied with an input and measured, preserves the ambiguity and allows both compatible network matrices to have spectral radius, the largest eigenvalue magnitude, below any prescribed positive bound. The resulting structure has twelve nodes, forty-two independent unknown coefficients, and zero diagonal. Ambiguity persists on a nonempty open set of the full parameter space, with both candidates satisfying the same spectral-radius bound. An explicit rational pair has exact matrix-power bounds. The constant coefficients also define proper transfer functions: rational functions whose numerator degree does not exceed their denominator degree. Replacing scalar coefficients by rational transfer functions preserves the two-solution construction.**

***Index Terms*— Identification, network analysis and control, linear systems.**

I. Introduction [1]

A linear network consists of node signals related by linear equations along prescribed directed edges. An external input excites a node when it enters that node's equation; a measurement records a selected node signal. Network identification seeks to recover the unknown edge coefficients from the response of the measured signals to the external inputs. When only some nodes are excited or measured, distinct choices of internal coefficients can produce the same observed response.

Local identifiability means that the coefficients are unique among sufficiently close admissible choices. Global identifiability requires uniqueness among all admissible choices. For scalar coefficients, a property is called generic if it holds except on a set of parameter values of zero volume. The distinction matters when a test based on derivatives establishes local uniqueness: such a test need not exclude a second, isolated explanation of the same input-output relation.

Legat and Hendrickx developed a generic local identifiability test for partial excitation and measurement and asked whether local and global identifiability can differ in this setting [2]. The two notions coincide when all nodes are excited or all nodes are measured [3]. A further equivalence result holds for a specified class of two-part networks with known internal coefficients [4]. The general question remains open in the discussion of [3, Sec. II-C] and [5, Sec. 12.2]. The latter reports an unsuccessful search for a counterexample with up to five unknown edges, subject to specified structural restrictions.

This letter gives a negative answer for networks with independently unknown complex scalar coefficients and fixed selections of excited and measured nodes. First, a six-node construction has exactly two compatible coefficient vectors generically. Second, adding excited and measured nodes preserves the ambiguity while controlling the spectral radius, the largest magnitude of a matrix eigenvalue. Both alternatives satisfy the restriction on this quantity imposed in [3]. The augmented structure has twelve nodes and forty-two independently unknown coefficients. For every positive bound on the spectral radius, ambiguity occurs on a nonempty open subset of the full parameter space of forty-two complex coefficients.

Section II states the admissible classes and the precise identifiability definitions. Sections III and IV develop the two constructions. Section V proves the spectral-radius result and gives an explicit rational pair. Section VI extends the result to rational transfer functions under an algebraic definition of generic identifiability.

II. Network Description and Identifiability [1]

A transfer function describes a linear relation between signals in the frequency domain. We follow the scalar formulation of [2, Secs. I and III] and [3, Sec. III-B]: each edge coefficient represents a transfer-function value at a fixed frequency and can therefore be complex.

Let n be the number of nodes, numbered $1, \dots, n$, and let E be a prescribed set of ordered pairs (i, j), where $i, j \in \{1, \dots, n\}$ and $i \neq j$. An edge (i, j) carries the signal from node j to node i. Write $m = |E|$, fix an ordering of these edges, and collect their coefficients in a vector $\theta \in \mathbb{C}^m$. The network matrix $G(\theta) \in \mathbb{C}^{n \times n}$ places these coefficients in the positions in E and has zeros elsewhere. Each edge has its own independent parameter. In particular, $G(\theta)$ has zero diagonal.

Let n_B and n_C be the numbers of excited and measured nodes. The fixed matrix $B \in \{0,1\}^{n \times n_B}$ has distinct columns of the identity matrix, and $C \in \{0,1\}^{n_C \times n}$ has distinct rows of the identity matrix. Thus B inserts inputs at the excited nodes and C selects the measured signals. At the chosen frequency, let $w \in \mathbb{C}^n$, $r \in \mathbb{C}^{n_B}$, and $y \in \mathbb{C}^{n_C}$ be the complex amplitudes of the node signals, inputs, and

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measurements, respectively. The network equations are

$$w = G(\theta)w + Br, \qquad y = Cw. \tag{1}$$

For a positive integer d, let I_d denote the $d \times d$ identity matrix. We call (1) well posed if $I_n - G(\theta)$ is invertible. In that case the transfer matrix from the external inputs r to the measurements y, evaluated at the chosen frequency, is

$$\Phi(\theta) = C(I_n - G(\theta))^{-1}B. \tag{2}$$

Identification concerns recovery of θ from $\Phi(\theta)$.

Define the set of well-posed parameters by

$$\Omega = \{\theta \in \mathbb{C}^m : \det(I_n - G(\theta)) \neq 0\}.$$

For a matrix, let ρ denote its spectral radius. For $0 < \varepsilon \leq 1$, define the more restricted set

$$\Omega_\varepsilon = \{\theta \in \mathbb{C}^m : \rho(G(\theta)) < \varepsilon\}. \tag{3}$$

The strict inequality implies $\Omega_\varepsilon \subseteq \Omega$. Openness and volume are understood by identifying $\mathbb{C}^m$ with $\mathbb{R}^{2m}$ through real and imaginary parts. Both parameter sets are open in this topology.

Let $\mathcal{D}$ be either Ω or Ω_ε. The network is *locally identifiable at $\theta \in \mathcal{D}$* if some neighborhood N of θ satisfies the following implication for every alternative vector $\widetilde{\theta}$:

$$\widetilde{\theta} \in N \cap \mathcal{D}, \quad \Phi(\widetilde{\theta}) = \Phi(\theta) \quad \Longrightarrow \quad \widetilde{\theta} = \theta.$$

It is *globally identifiable at θ in $\mathcal{D}$* if the same implication holds for every $\widetilde{\theta} \in \mathcal{D}$. Thus any competing network must satisfy the same admissibility restriction.

The structure is generically locally or globally identifiable in $\mathcal{D}$ if the corresponding property holds outside a subset of $\mathcal{D}$ of Lebesgue measure zero, that is, zero volume. We use the complex Jacobian, the matrix of derivatives of the entries of Φ with respect to the complex coordinates of θ. All ranks are over $\mathbb{C}$. We prove generic local identifiability by showing that this Jacobian has column rank m except where a nonzero polynomial vanishes. Such an exceptional set has zero volume in $\mathbb{R}^{2m}$. For the rational map Φ, complex rank m gives real rank $2m$ after separating real and imaginary parts. Applying the inverse function theorem to m suitable complex entries of Φ then gives local uniqueness. Conversely, a nonempty open set of parameters with distinct admissible alternatives suffices to disprove generic global identifiability.

The generic claims in Sections III–V refer to the independent scalar parameters. Section VI considers independent rational transfer functions as the unknowns.

III. A Six-Node Ambiguity [1]

Consider nine independent complex parameters a_i, d_i, x_i, indexed by $i \in \{1, 2, 3\}$. In this section and in formulas using these parameters, subscripts are cyclic: $i+1 = 1$ when $i = 3$, and $i - 1 = 3$ when $i = 1$. Let diag form a diagonal matrix from its arguments, and define

$$A = \begin{pmatrix} 0 & 0 & a_3 \\ a_1 & 0 & 0 \\ 0 & a_2 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & d_1 & 0 \\ 0 & 0 & d_2 \\ d_3 & 0 & 0 \end{pmatrix}, \tag{4}$$
$$X = \operatorname{diag}(x_1, x_2, x_3),$$

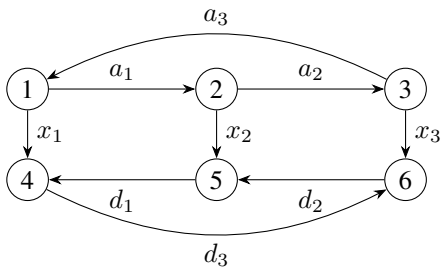


Fig. 1. The six-node construction. Nodes $1, 2, 3$ are excited; nodes $4, 5, 6$ are measured. Each labeled edge has an independent unknown coefficient.

In block matrices, zero blocks have the dimensions determined by their positions. Set

$$G_b = \begin{pmatrix} A & 0 \\ X & D \end{pmatrix}, \quad B_b = \begin{pmatrix} I_3 \\ 0 \end{pmatrix}, \quad C_b = \begin{pmatrix} 0 & I_3 \end{pmatrix}. \tag{5}$$

This is (1) with six nodes, the first three excited and the last three measured. Figure 1 shows the edges. Define the products $\alpha = a_1a_2a_3$ and $\delta = d_1d_2d_3$. Then

$$\det(I_6 - G_b) = (1 - \alpha)(1 - \delta).$$

For well-posed parameters, denote the transfer matrix by

$$F = C_b(I_6 - G_b)^{-1}B_b = (I_3 - D)^{-1}X(I_3 - A)^{-1}. \tag{6}$$

Proposition 1: The structure (5) is generically locally identifiable in Ω. For generic parameters, exactly two well-posed complex parameter vectors give the same transfer matrix F.

Proof: Assume first that $x_1x_2x_3 \neq 0$, so that F is invertible. Introduce coordinates $u_i = a_i/x_i$, $v_i = d_i/x_i$, and $z_i = 1/x_i$. This change is invertible, with $a_i = u_i/z_i$, $d_i = v_i/z_i$, and $x_i = 1/z_i$. The inverse transfer matrix is

$$\begin{aligned} F^{-1} &= (I_3 - A)X^{-1}(I_3 - D) \\ &= \begin{pmatrix} z_1 + u_3v_3/z_3 & -v_1 & -u_3 \\ -u_1 & z_2 + u_1v_1/z_1 & -v_2 \\ -v_3 & -u_2 & z_3 + u_2v_2/z_2 \end{pmatrix}. \end{aligned} \tag{7}$$

Thus the off-diagonal entries determine u_i, v_i. Let $e_i = (F^{-1})_{ii}$ be the diagonal entries. They satisfy

$$e_i = z_i + \frac{u_{i-1}v_{i-1}}{z_{i-1}}. \tag{8}$$

With u_i, v_i fixed, the Jacobian of these three equations with respect to (z_1, z_2, z_3), denoted by J_z, has determinant

$$\det J_z = 1 - \frac{u_1v_1u_2v_2u_3v_3}{z_1^2z_2^2z_3^2} = 1 - \alpha\delta. \tag{9}$$

The six directly determined coordinates and (9) give Jacobian rank nine when $\alpha\delta \neq 1$. Inverting an invertible transfer matrix preserves rank. This proves generic local identifiability.

To construct a second solution, define

$$\begin{aligned} k_i &= a_id_i\frac{x_{i+1}}{x_i}, \qquad h_i = 1 + k_i + k_ik_{i+1}, \\ \eta_i &= \frac{k_ih_{i+1}}{h_i}, \qquad (a_i', d_i', x_i') = \frac{1}{\eta_i}(a_i, d_i, x_i). \end{aligned} \tag{10}$$

Primes denote the alternative parameter values. The following identities follow by expanding h_i and using $k_1k_2k_3 = \alpha\delta$:

$$\begin{aligned} \eta_i - 1 &= \frac{\alpha\delta - 1}{h_i}, \\ \eta_i + \frac{k_{i-1}}{\eta_{i-1}} &= 1 + k_{i-1}, \qquad \eta_1\eta_2\eta_3 = \alpha\delta. \end{aligned} \tag{11}$$

The transformation fixes u_i, v_i and gives $z_i' = \eta_i z_i$. Since $u_{i-1}v_{i-1}/z_{i-1} = k_{i-1}z_i$, the second identity in (11) preserves (8), and hence F. The first identity makes the alternatives distinct when $\alpha\delta \neq 1$. Denote the products of the primed cycle coefficients by $\alpha' = \prod_{i=1}^3 a_i'$ and $\delta' = \prod_{i=1}^3 d_i'$. The last identity gives

$$\alpha' = \frac{1}{\delta}, \qquad \delta' = \frac{1}{\alpha}. \tag{12}$$

Consequently the alternative is well posed whenever $\alpha\delta \neq 0$ and $\alpha, \delta \neq 1$.

It remains to bound the number of solutions. Define $b_i = u_iv_i$. Equation (8) gives successively

$$z_2 = e_2 - b_1/z_1, \quad z_3 = e_3 - b_2/z_2, \quad z_1 = e_1 - b_3/z_3.$$

Substitution and multiplication by the denominators yield

$$\begin{aligned} 0 = {} & (b_2 - e_2e_3)z_1^2 \\ & + (b_1e_3 - b_2e_1 - b_3e_2 + e_1e_2e_3)z_1 \\ & + b_1b_3 - b_1e_1e_3. \end{aligned} \tag{13}$$

Generically its leading coefficient is nonzero, so there are at most two values of z_1. Each determines z_2, z_3 and all original parameters. A candidate with some $x_i = 0$ cannot yield an invertible F, by (6). The rational formulas (10) supply a distinct complex second solution wherever the stated denominators are nonzero and $\alpha\delta \neq 1$. These exclusions are zeros of nonzero polynomials. For example, $a_i = d_i = -1/2$ and $x_i = 1$ satisfy all of them. This proves the assertion about exactly two solutions. ∎

The six-node construction alone does not give two alternatives in Ω_1. Indeed, its block triangular form gives

$$\rho(G_b) = \max\{|\alpha|^{1/3}, |\delta|^{1/3}\}.$$

If both products have magnitude below one, (12) makes the alternative violate the same restriction. The next construction changes the full network eigenvalues while retaining the ambiguity.

IV. Adding Excited and Measured Nodes [1]

Let $w_b \in \mathbb{C}^6$ collect the original node amplitudes. Add six nodes with amplitudes $w_c \in \mathbb{C}^6$, all excited and measured. Let $K \in \mathbb{C}^{3\times 6}$, $L_c \in \mathbb{C}^{6\times 3}$, and $J \in \mathbb{C}^{6\times 6}$ specify the additional connections. Define the augmented network and its selections by

$$\begin{aligned} \mathcal{G} &= \begin{pmatrix} G_b & B_bK \\ L_cC_b & J \end{pmatrix}, \\ \mathcal{B} &= \begin{pmatrix} B_b & 0 \\ 0 & I_6 \end{pmatrix}, \qquad \mathcal{C} = \begin{pmatrix} C_b & 0 \\ 0 & I_6 \end{pmatrix}. \end{aligned} \tag{14}$$

This again has the form (1); its twelve node amplitudes are the six original amplitudes w_b followed by w_c. The matrices K, L_c, J connect the added nodes to the original excited and measured nodes and to each other. Denote its transfer matrix by

$$H = \mathcal{C}(I_{12} - \mathcal{G})^{-1}\mathcal{B}. \tag{15}$$

Lemma 1: Suppose $I_6 - G_b$, X, and $I_{12} - \mathcal{G}$ are invertible. Then H is invertible and

$$H^{-1} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix}. \tag{16}$$

Consequently H determines each entry of K, L_c, J, and, on this domain, equality of augmented transfer matrices is equivalent to equality of F, K, L_c, J.

Proof: Let $r_b \in \mathbb{C}^3$ and $s \in \mathbb{C}^6$ be the input amplitudes at the original and added nodes, respectively, and let $y_b \in \mathbb{C}^3$ be the original measured amplitudes. The node equations give

$$y_b = F(r_b + Kw_c), \qquad w_c = L_cy_b + Jw_c + s.$$

Thus

$$\begin{pmatrix} r_b \\ s \end{pmatrix} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix} \begin{pmatrix} y_b \\ w_c \end{pmatrix}.$$

By well-posedness, (y_b, w_c) is obtained from (r_b, s) through the square matrix H. The displayed matrix is therefore a left inverse of H, proving (16) and the remaining claims. ∎

This identity also preserves the rank information needed for local identifiability. If q_c entries of K, L_c, J are independently unknown and the remaining entries are fixed zeros, their derivatives occupy positions in (16) disjoint from those of F^{-1}. The augmented Jacobian rank is therefore $q_c + 9$ wherever the base Jacobian has rank nine. The choice below makes both compatible augmented matrices satisfy the spectral-radius restriction.

V. Ambiguity With Arbitrarily Small Spectral Radius [1]

Use superscript T for transpose. Define the cyclic permutation matrix U, and then V, by

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad V = U + U^T.$$

These matrices satisfy $U^3 = I_3$ and $U^T = U^2$. For real scalars $t > 0$ and $x \neq 0$, define

$$M(t, x) = \begin{pmatrix} -tU & 0 & V & -U \\ xI_3 & -tU^T & 0 & 0 \\ 0 & U^T & U^T & -V \\ 0 & -V & I_3 & U \end{pmatrix}. \tag{17}$$

All blocks in this display have size 3×3. Use the nonzero positions in $M(t, x)$ as the permitted edge set for $\mathcal{G}$. This pattern does not depend on the chosen nonzero values of t, x. Assign a separate independent unknown coefficient to every permitted position. There are nine in G_b, nine in K, nine in L_c, and fifteen in J, giving $m = 42$. Thus the repeated

numerical values in (17) specify particular parameter choices in a class with forty-two independent complex coefficients.

The selections $\mathcal{B},\mathcal{C}$ in (14) excite nodes 1–3 and 7–12 and measure nodes 4–12. There are no diagonal edges. Let $\theta \in \mathbb{C}^{42}$ collect the independent coefficients of this structure, and apply (3) to $\mathcal{G}(\theta)$.

Theorem 1: For every $0 < \varepsilon \leq 1$, this twelve-node structure is generically locally identifiable in Ω_ε. There is a nonempty open set $W_\varepsilon \subset \Omega_\varepsilon$ such that every $\theta \in W_\varepsilon$ has a distinct alternative $\widetilde{\theta} \in \Omega_\varepsilon$ with the same transfer matrix H. Both alternatives are locally identifiable. Consequently, the structure is not generically globally identifiable in Ω_ε.

Proof: We first give equal transfer matrices, then make both spectral radii small, and finally establish ambiguity on an open set of all forty-two parameters.

Equal transfer matrices: The matrices in (17) use the fixed numerical choices

$$K_0 = \begin{pmatrix} V & -U \end{pmatrix}, \quad L_0 = \begin{pmatrix} U^T \\ -V \end{pmatrix}, \quad J_0 = \begin{pmatrix} U^T & -V \\ I_3 & U \end{pmatrix}. \tag{18}$$

For their base block, $A = -tU$, $D = -tU^T$, and $X = xI_3$. Write $F(t,x)$ for the corresponding value of (6). Since $t > 0$, the base block is well posed, and

$$F(t,x)^{-1} = x^{-1}(I_3{+}tU)(I_3{+}tU^T) = x^{-1}\big[(1{+}t^2)I_3{+}tV\big]. \tag{19}$$

Hence $F(t,1) = F(t^{-1},t^{-2})$. The two augmented matrices

$$\mathcal{G}_1(t) = M(t,1), \qquad \mathcal{G}_2(t) = M(t^{-1},t^{-2}) \tag{20}$$

use the same K_0, L_0, J_0. By Lemma 1, their transfer matrices agree whenever both augmented networks are well posed. They are distinct for $t \neq 1$.

Small spectral radii: At $t = 1$, both matrices in (20) equal $M(1,1)$. Let G_{b0} be its base block. Direct substitution gives

$$J_0 = G_{b0} + B_b K_0 - L_0 C_b.$$

Define the two matrices

$$\begin{aligned} E_+ &= G_{b0} + B_b K_0 = \begin{pmatrix} U^T & -U \\ I_3 & -U^T \end{pmatrix}, \\ E_- &= G_{b0} - L_0 C_b = \begin{pmatrix} -U & -U^T \\ I_3 & U \end{pmatrix}. \end{aligned} \tag{21}$$

Changing coordinates from (w_b, w_c) to $(w_b, w_b - w_c)$ makes $M(1,1)$ similar to the block upper triangular matrix

$$\begin{pmatrix} E_+ & -B_b K_0 \\ 0 & E_- \end{pmatrix}. \tag{22}$$

The identities for U imply $E_+^2 = E_-^2 = 0$. Squaring (22) therefore gives a matrix whose only possibly nonzero block is the upper right block; its square is zero. Consequently

$$M(1,1)^4 = 0. \tag{23}$$

For a $d \times d$ matrix Q, let $\|Q\|_\infty = \max_{1\leq i\leq d} \sum_{j=1}^{d} |Q_{ij}|$, its maximum absolute row sum. The standard eigenvalue bound gives

$$\rho(Q)^4 = \rho(Q^4) \leq \|Q^4\|_\infty. \tag{24}$$

Both matrices in (20) converge to $M(1,1)$ as $t \to 1$. Their fourth-power norms consequently tend to zero. For any prescribed $0 < \varepsilon \leq 1$, choose $t \neq 1$ sufficiently close to one that both norms are below ε^4. Equation (24) then places both distinct matrices in Ω_ε, so their transfer matrices are well defined and equal.

Local identifiability: The thirty-three independent added coefficients are determined directly in (16). The base Jacobian has rank nine by (9), since $\alpha\delta = t^6$ at the first point of (20) and $\alpha\delta = t^{-6}$ at the second. Both augmented Jacobians therefore have rank $33{+}9 = 42$. All their entries are rational functions of the forty-two parameters. A minor, meaning the determinant of a square submatrix, that is nonzero at one of these points has a numerator that is a nonzero polynomial. Outside its zero set, the Jacobian of H has full column rank. This establishes generic local identifiability in Ω_ε.

An open set of ambiguous parameters: Define $\mathcal{T}$ on the forty-two parameters by applying (10) to the nine base coefficients and leaving the thirty-three added coefficients unchanged. At the first point of (20), one has $k_i = t^2$, $h_i = 1 + t^2 + t^4$, and $\eta_i = t^2$; thus $\mathcal{T}$ maps it to the second point. All denominators are nonzero there. The transformation is rational and continuous in a neighborhood, and preserves H by Lemma 1.

At these two points the fourth-power bounds are strict, the alternatives are distinct, and both Jacobians have full column rank. These properties persist in a sufficiently small open neighborhood W_ε of the first point and its image under $\mathcal{T}$. Shrink the neighborhood further, if necessary, so that the base inverses in the lemma exist and all permitted coefficients remain nonzero. Then every $\theta \in W_\varepsilon$ and its alternative $\mathcal{T}(\theta)$ belong to Ω_ε, have equal transfer matrices, and are distinct and locally identifiable. The neighborhood is open in $\mathbb{C}^{42}$, so it has positive volume in the real and imaginary coordinates of $\mathbb{R}^{84}$. This disproves generic global identifiability. ∎

The conclusion also holds when all edge coefficients are restricted to be real. The explicit pair in (20) is real, the transformation $\mathcal{T}$ preserves real coefficients, and its neighborhood contains an open set of real parameters. The same rank argument proves generic local identifiability on that real parameter domain.

A. An explicit rational pair

For $\varepsilon = 1$, choose $t = 99/100$ in (20). The common inverse transfer matrix, obtained from (16), is

$$H^{-1} = \begin{pmatrix} (1+t^2)I_3 + tV & -V & U \\ -U^T & I_3 - U^T & V \\ V & -I_3 & I_3 - U \end{pmatrix}. \tag{25}$$

Exact multiplication of the rational matrices gives

$$\begin{aligned} \|\mathcal{G}_1(99/100)^4\|_\infty &= \frac{33910001}{100000000} < 1, \\ \|\mathcal{G}_2(99/100)^4\|_\infty &= \frac{16976126}{32019867} < 1. \end{aligned} \tag{26}$$

These inequalities certify admissibility through (24). Their spectral radii are approximately 0.509563 and 0.404210, respectively. The exact equalities (25) and (26) make the

pair directly checkable with rational arithmetic. Theorem 1 establishes the open-set conclusion independently of these numerical values.

VI. Transfer Functions and Conclusions [1]

A. Algebraic identifiability over rational functions

Let λ be a symbolic complex frequency variable and let $\mathbb{K} = \mathbb{C}(\lambda)$ be the field of ratios of polynomials in λ with complex coefficients. We now replace the scalar edge coefficients by elements of $\mathbb{K}$, keeping the same network structures and formulas for the transfer matrices. Each coordinate of $\theta \in \mathbb{K}^m$ is then a rational function of λ. Equality of transfer matrices means an identity of rational functions.

We use the algebraic framework of [6, Sec. V]. For the algebraic calculation, the edge coordinates are independent variables over $\mathbb{K}$. Solutions are counted in an algebraic closure $\overline{\mathbb{K}}$, a field containing $\mathbb{K}$ in which every nonconstant polynomial has a root. An algebraic set is a set defined by polynomial equations in these coordinates. An algebraically generic property holds outside a proper algebraic subset, meaning one smaller than the full parameter domain or the image of the map $\theta \mapsto \Phi(\theta)$, as appropriate. The fiber at a transfer matrix is the set of admissible parameter vectors producing it. Generic local identifiability in the algebraic sense means that the generic fiber has dimension zero, equivalently that it consists of finitely many isolated points. Generic global identifiability requires one point. Admissibility here excludes zeros of the determinants and denominators required by the network equations.

In the notation of [6], the matrices P, Q_r, R_y, S_{yr} define the closed-loop transfer function $R_y P^{-1} Q_r + S_{yr}$ from external inputs to measured outputs. Our augmented network is its specialization

$$P = I_{12} - \mathcal{G}, \qquad Q_r = \mathcal{B}, \qquad R_y = \mathcal{C}, \qquad S_{yr} = 0.$$

Proposition 2: Over $\mathbb{K}$, the six-node structure (5) and the twelve-node structure of Theorem 1 each have exactly two points in the generic algebraic fiber. Both are therefore generically locally identifiable in the algebraic sense and fail generic global identifiability. If one generic parameter vector has coordinates in $\mathbb{K}$, its distinct alternative also has coordinates in $\mathbb{K}$.

Proof: All identities in Proposition 1 use only field operations and remain valid over $\overline{\mathbb{K}}$. The quadratic (13) permits at most two base solutions. Formula (10) provides a distinct second solution outside the stated algebraic exclusions. Because this formula is rational, it maps $\mathbb{K}$-valued parameters to $\mathbb{K}$-valued parameters.

For a generic transfer matrix H, both H and the upper left 3×3 block of H^{-1} are invertible. Any well-posed candidate producing this H must have invertible X. Otherwise choose a nonzero vector $\xi \in \overline{\mathbb{K}}^3$ with $X\xi = 0$. The node amplitudes $(\xi, 0, 0)$, partitioned into groups of sizes three, three, and six, give zero output for input $((I_3 - A)\xi, 0)$. Well-posedness makes this input nonzero, contradicting invertibility of H. Eliminating the first three node amplitudes now gives

$$H^{-1} = \begin{pmatrix} (I_3 - A)X^{-1}(I_3 - D) & -K \\ -L_c & I_6 - J \end{pmatrix}.$$

Invertibility of its upper left block makes $I_3 - A$ and $I_3 - D$ invertible. Hence every candidate satisfies the lemma's assumptions. The added coefficients are determined, and the remaining fiber is exactly the two base solutions. For either base solution, the displayed invertible matrix determines the measured node amplitudes from the inputs, and $X^{-1}(I_3 - D)$ determines the first three node amplitudes. Thus both augmented solutions are well posed. The transformation $\mathcal{T}$ leaves the added coefficients unchanged, so it also preserves $\mathbb{K}$-valued parameters. ∎

B. Properness and spectral-radius restrictions

A rational transfer function is *proper* if its numerator degree does not exceed its denominator degree, and *strictly proper* if the inequality is strict. Each real coefficient in (20) defines a constant proper rational transfer function. The matrix $(I_{12} - \mathcal{G})^{-1}$, which maps additive inputs at all nodes to the node signals, is also constant. It maps bounded inputs to bounded signals, so the network is stable in this input-output sense. Both networks have identical transfer matrices and satisfy the spectral-radius restriction at every frequency.

Algebraically generic uniqueness also fails after imposing properness, stability, and the spectral-radius bound. Indeed, $W_\varepsilon \cap \mathbb{R}^{42}$ from Theorem 1 contains a nonempty real open set of ambiguous constant networks. This set is Zariski dense over $\overline{\mathbb{K}}$, meaning that no nonzero polynomial in the forty-two edge coordinates vanishes on all of it. To see this, take a real open box contained in the set. Fix all but one coordinate in their respective intervals. A polynomial vanishing throughout the box has infinitely many roots in the remaining coordinate, so each of its coefficients is zero. Repeating this argument for the other coordinates proves that the polynomial is zero. Thus no proper algebraic exceptional set can contain all these admissible ambiguous networks. Their transfer matrices are also Zariski dense in the image of the map $\theta \mapsto H(\theta)$. Indeed, suppose a polynomial vanishes on all these transfer matrices but not on the full image. Substituting $H(\theta)$ and clearing denominators gives a nonzero polynomial that vanishes on the real open set, a contradiction.

This conclusion uses the edge transfer functions themselves as coordinates over $\mathbb{K}$. A separate analysis is required when genericity refers to the numerical coefficients of a prescribed positive-order transfer-function family, or when the admissible networks must be strictly proper.

The construction separates local recovery from global uniqueness even under an arbitrarily small positive spectral-radius bound. Its mechanism is explicit: the six-node equations admit two isolated solutions, and the added excited and measured nodes preserve those solutions while moving all eigenvalues of both complete networks near zero. Thus a derivative rank test for local identifiability cannot establish generic global identifiability for this class of partially excited and measured networks.

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REFERENCES

[1] OpenAI, "Codex," software. [Online]. Available: https://openai.com/codex/. Accessed: Oct. 8, 2026.

[2] A. Legat and J. M. Hendrickx, "Local network identifiability with partial excitation and measurement," in *Proc. 59th IEEE Conf. Decision Control*, 2020, pp. 4342–4347, doi: 10.1109/CDC42340.2020.9303890.

[3] A. Legat and J. M. Hendrickx, "Identifiability in networked systems: Algebraic and combinatorial conditions," manuscript, 2024. [Online]. Available: https://perso.uclouvain.be/antoine.legat/_static/TAC24.pdf.

[4] A. Legat and J. M. Hendrickx, "Combinatorial characterization for global identifiability of separable networks with partial excitation and measurement," in *Proc. 62nd IEEE Conf. Decision Control*, 2023, pp. 2471–2476, doi: 10.1109/CDC49753.2023.10383708.

[5] A. Legat, "Generic local identifiability in dynamical networks: From algebraic characterization to algorithms for signal selection," Ph.D. dissertation, UCLouvain, Louvain-la-Neuve, Belgium, 2025. [Online]. Available: https://hdl.handle.net/2078.5/256546.

[6] A. Hansson, J. V. Galvão da Mata, and M. S. Andersen, "Informativity and identifiability for identification of networks of dynamical systems," arXiv:2602.20107, 2026. [Online]. Available: https://arxiv.org/abs/2602.20107v3.